

Strategy Audit Report

Sample Backtest Analysis

Strategy Overview

- **Asset:** SPY (S&P 500 ETF)
- **Timeframe:** Daily
- **Strategy Type:** Machine Learning (LightGBM)
- **Period:** Jan 2020 - Dec 2025

Performance Metrics

Metric	Value	Benchmark
Sharpe Ratio	1.16	> 1.0 ✓
Total Return	+47.3%	S&P: +42.1%
Max Drawdown	-7.0%	S&P: -33.9%
Win Rate	58.2%	> 50% ✓
Profit Factor	1.82	> 1.5 ✓

Risk Analysis

Risk Metric	Value	Status
Sortino Ratio	1.84	Excellent
Calmar Ratio	2.31	Excellent
Value at Risk (95%)	-2.1%	Low Risk
Feature Exposure	0.26	Good

Monte Carlo Simulation (1000 paths)

- **Median Final Equity:** \$147,320
- **5th Percentile:** \$118,450
- **95th Percentile:** \$189,230
- **Probability of Profit:** 94.2%

Validation

- ✓ Walk-Forward Optimization: 8 folds, all profitable
 - ✓ Out-of-sample testing: Consistent performance
 - ✓ Live paper trading: Verified with Alpaca
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Recommendations

1. Strategy is **production-ready** for paper trading
 2. Recommended position sizing: 1-2% risk per trade
 3. Suggested initial capital: \$10,000+
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*Report generated by Professional Trading Bot Framework
Python + LightGBM + Alpaca/MetaTrader 5*